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(54) **SECURITY SYSTEM WITH CLOUD-BASED CONTROLLER**

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G16Y 40/10; G16Y 40/50

See application file for complete search history.

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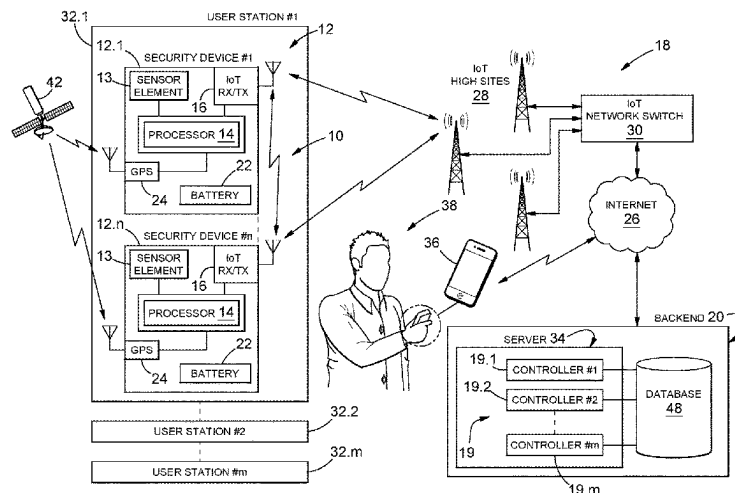
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(57) **ABSTRACT**

The self-contained security device **12.1** comprises a sensor element **13** and a processor **14** connected to the sensor element for generating an intrusion message in response to a signal from the sensor element. An Internet of things (IoT) transmitter **16** is connected to the processor **14** for transmitting to a cloud-based controller **19** the intrusion message via an Internet of things communications path **18** which is provided by an Internet of things network.

12 Claims, 3 Drawing Sheets



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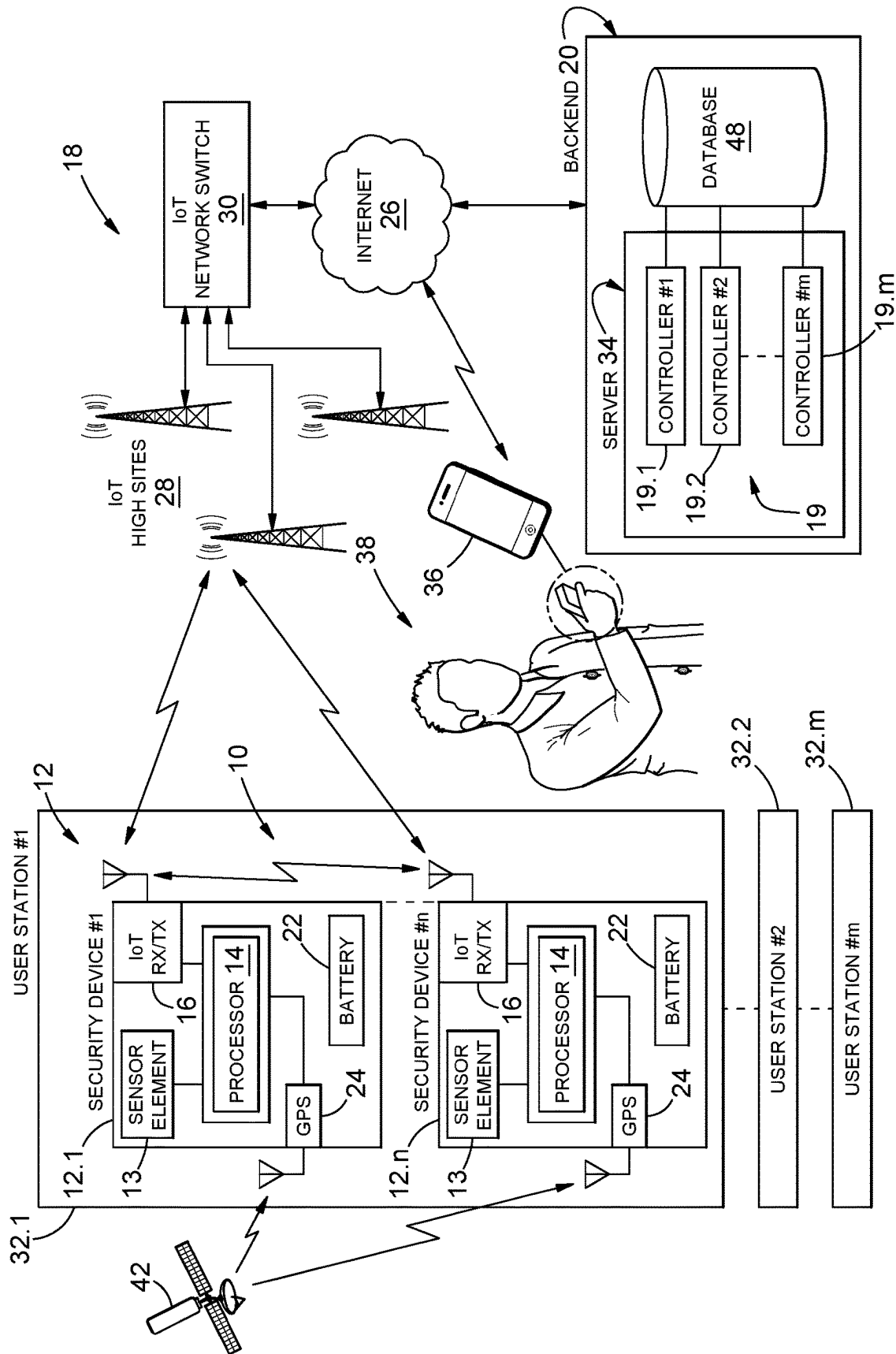


FIGURE 1

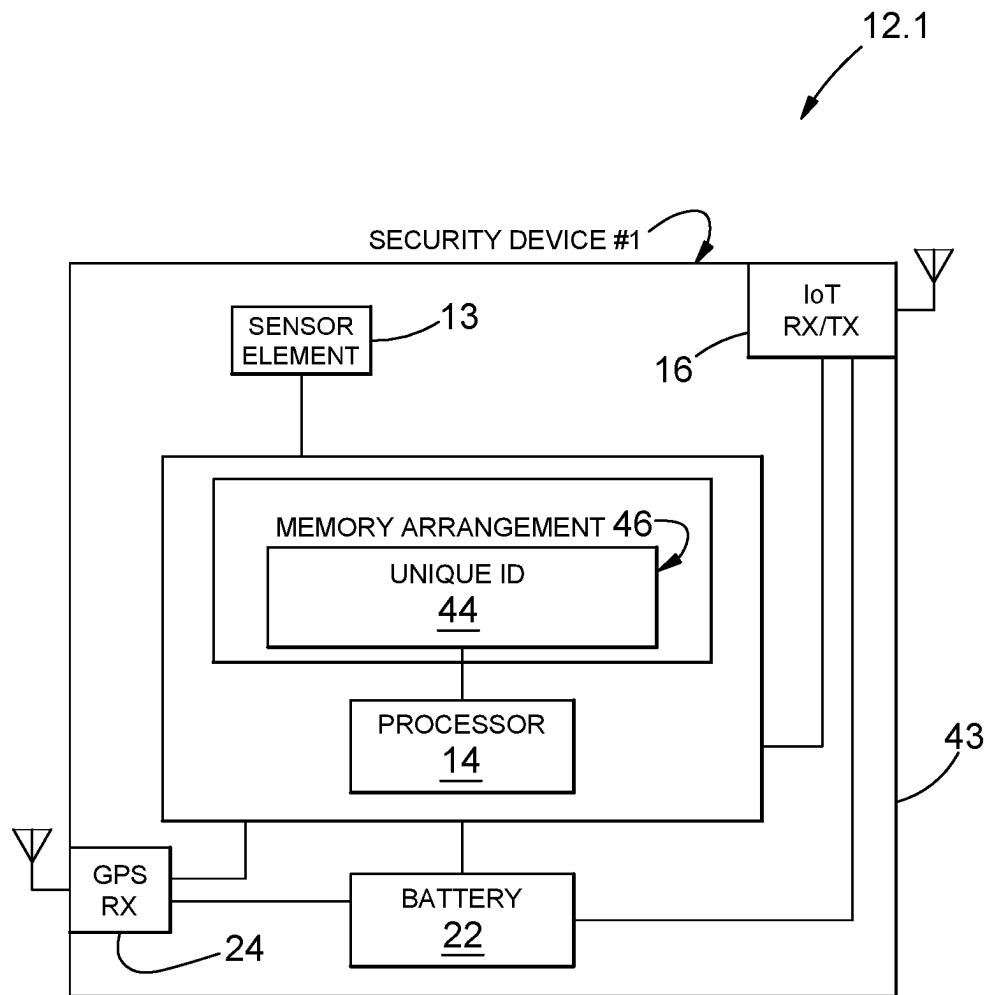
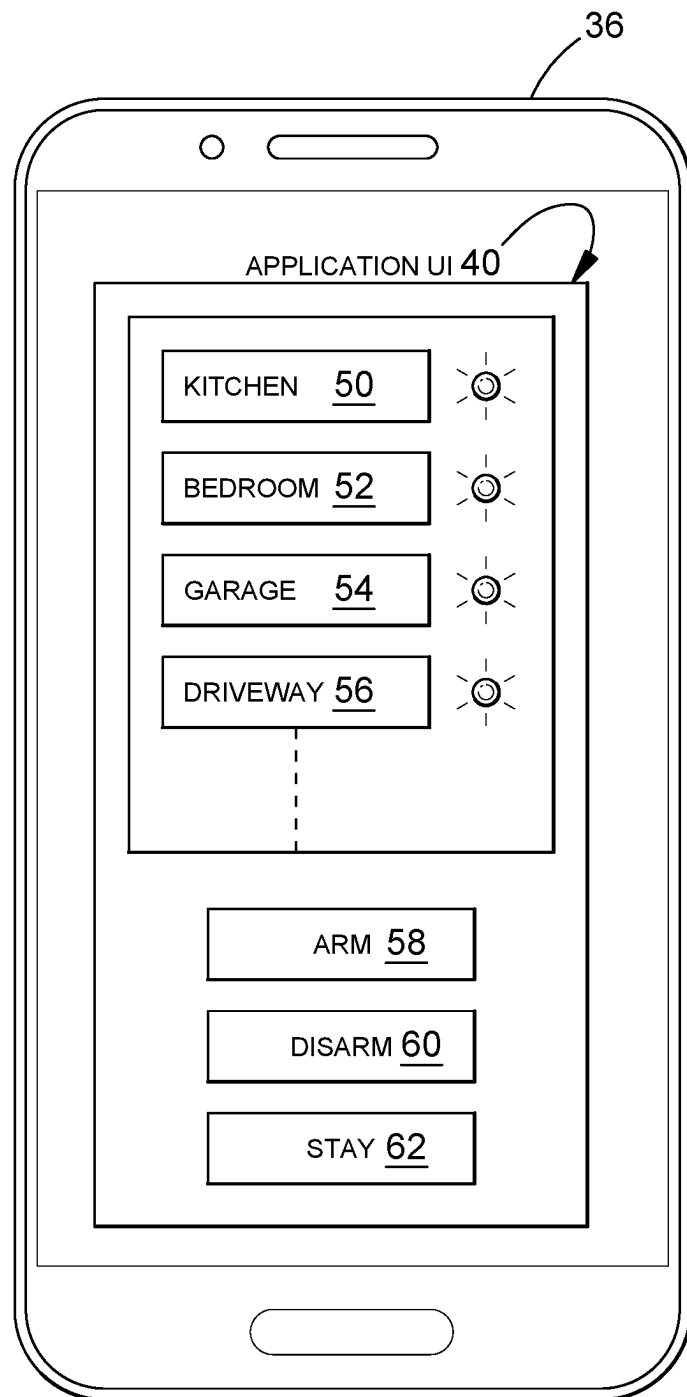


FIGURE 2

FIGURE 3

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SECURITY SYSTEM WITH CLOUD-BASED CONTROLLER

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a National Stage Application, filed under 35 U.S.C. § 371, of International Application No. PCT/IB2018/059816, filed Dec. 10, 2018, which claims priority to South Africa Application No. 2017/08671, filed Dec. 20, 2017; the contents of both of which are hereby incorporated by reference in their entirety.

BACKGROUND

Related Field

DESCRIPTION OF RELATED ART

This invention relates to a security device, a security system and to a method of protecting a user station.

Security systems are generally used at user stations, such as residential and commercial properties, to detect unauthorised entry at the station and to alert a security service provider, normally operating from a central backend, thereby to protect property at the station from theft as well as to provide personal protection.

The known security systems utilise a local controller at the station. The controller comprises a plurality of inputs which are connected to a plurality of intrusion sensors which are distributed at the station. When one of the intrusion sensors senses an intrusion, an alarm is activated by the controller. An intrusion message is transmitted by the local controller from the user station to the remote central backend via a conventional telephone line utilising a connection typically referred to as a “tip and ring” connection. Alternatively, a Global System for Mobile Communications (GSM) modem is utilised to transmit the intrusion message to the backend. These GSM modems may be jammed by intruders and the telephone line may be compromised.

Furthermore, these local controllers and associated equipment need to be installed and programmed at the user station. This installation process is technical and therefore a skilled technician or relevant expert is generally required to perform the installation, which may be both inconvenient to and expensive for the user. Hence, these known security systems may not be suitable for at least some applications.

BRIEF SUMMARY

Accordingly, it is an object of the present invention to provide a security device, a security system and a method of protecting a user station with which the applicant believes the aforementioned problems may at least be alleviated or which may provide a useful alternative for the known systems and/or devices.

According to a first aspect of the invention there is provided a security device comprising:

- a sensor element;
- a processor connected to the sensor element for generating a first message in response to a signal from the sensor element; and
- an Internet of things transmitter connected to the processor for transmitting to a cloud-based controller the first message via an Internet of things communications path provided by an Internet of things network.

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The Internet of things (IoT) is known in the art and comprises a network of things. These things typically comprise at least one of associated electronics, software, sensors and actuators. The things all have network connectivity, enabling them to connect and exchange data with one another and/or with a central backend over the Internet. Preferably, each thing forming part of the IoT network is uniquely identifiable in the IoT network. The IoT network typically comprises network dedicated IoT high sites.

Hence, the security device may be assigned a unique identification number (ID) for use in communications via the network.

The sensor element may comprise any one of a) an intrusion sensor, such as any one of: a passive infrared sensor, a contact sensor for a door or window, photoelectric beam, a panic button or other operable button, a magnetic lock sensor, a strike lock sensor; and b) another sensor, such as a temperature sensor, a smoke sensor, a water or moisture sensor/detector etc. These sensors may be normally open or normally closed sensors.

The first message may be an intrusion message comprising data derived from a signal from an intrusion sensor, alternatively the first message may be a warning message comprising data derived from a signal from at least one of the other sensors, such as a temperature sensor, a smoke sensor, a water or moisture sensor etc.

The security device may comprise a local power supply, preferably in the form of a replaceable battery, providing an output voltage. The battery may be rechargeable. The power supply is preferably dedicated and local to the security device.

The processor may be configured to incorporate, into the first message, data relating to a current output voltage of the battery

The security device may comprise a position determining device.

The position determining device may be configured to generate position data and the processor may be configured to incorporate in the first message the position data.

The security device may comprise an IoT receiver for receiving a second message, such as an instruction or command message, from the cloud-based controller via the IoT network.

The IoT transmitter and IoT receiver may be combined in an IoT transceiver.

At least the sensor element, processor, transmitter and power supply may be housed in a single housing for the security device. The housing may be dedicated to the security device. Preferably, all of the sensor element, processor, transmitter, receiver, position determining device and power supply are housed in the single housing.

The security device is hence self-contained and portable or transportable between different locations.

According to a second aspect of the invention there is provided a security system comprising:

at least one security device comprising:

- a sensor element;
 - a processor connected to the sensor element for generating a first message in response to a signal from the sensor element; and
 - an Internet of things transmitter connected to the processor for transmitting the first message via an Internet of things communications path; and
 - a cloud-based controller for the security system which is in data communication with the at least one security device via the Internet of things communications path.
- The at least one security device may be as defined above.

The cloud-based controller may be configured to perform at least some of the control functions which are conventionally associated with a local controller or control box of a conventional security system.

The security system may be associated with a first user station, the at least one security device may, in use, be deployed at the first user station and the cloud-based controller may be hosted at a remote central backend.

The cloud-based controller may be hosted on at least one server at the backend. The backend may be housed at a single physical site or may be distributed over a plurality of distributed physical sites.

The security system may comprise a terminal which is associated with the first user station, the terminal being configurable to be in communication with the cloud-based controller and may be user operable to configure settings of the cloud-based controller.

The terminal may be in the form of a mobile device, such as a smartphone, comprising a processor, an associated memory arrangement and a display.

The system may comprise a computer application program (app) which is stored in the memory arrangement and, in use, is executed by the processor to present a user interface (UI) on the display of the terminal, and in response to commands entered via the UI to configure the settings of or to program the cloud-based controller.

The system may comprise a plurality of the security devices, which may be located in distributed locations at the first user station.

Each of the plurality of security devices may be configured independently to communicate via its respective IoT transmitter or transceiver with the cloud-based controller.

Furthermore, any one of the plurality of security devices may be configured to communicate via its respective IoT transmitter or transceiver with any other one or more of the plurality of security devices.

The backend may host a respective cloud-based controller for each of a plurality of distributed user stations, each respective cloud-based controller forming part of a respective security system as defined above.

The backend may comprise a database which comprises data associated with the user stations, including data relating to the unique ID's of the security devices.

According to a third aspect of the invention there is provided a method of protecting a user station, the method comprising:

- at a cloud-based controller for a security system associated with the user station, receiving from a security device located at the user station an intrusion message via an Internet of things network; and
- causing the controller to act on the intrusion message in accordance with settings or rules which are pre-stored on the controller.

BRIEF DESCRIPTION OF THE FIGURES

The invention will now further be described, by way of example only, with reference to the accompanying diagrams wherein:

FIG. 1 is a high-level block diagram of an example embodiment of a security system;

FIG. 2 is a block diagram of an example embodiment of a security device forming part of the security system; and

FIG. 3 is a diagrammatic representation of a user interface (UI) presented by an application running on a mobile device forming part of the security system.

DETAILED DESCRIPTION OF VARIOUS EMBODIMENTS

An example embodiment of a security system is generally designated by the reference numeral **10** and an example embodiment of a self-contained security device is generally designated by the reference numeral **12.1** in FIGS. 1 and 2.

The self-contained security device **12.1** comprises a sensor element **13** and a processor **14** connected to the sensor element **13** for generating a first message in response to a signal from the sensor element **13**. An Internet of things (IoT) transmitter **16** is connected to the processor **14** for transmitting the first message to a cloud-based controller **19** via an IoT communications path **18** provided by an IoT network. The cloud-based controller **19** is preferably hosted at a remote central backend **20**. The security device **12.1** further comprises an IoT receiver. The IoT transmitter and the IoT receiver may be combined in an IoT transceiver **16**.

The self-contained security device **12.1** comprises a local power supply, preferably in the form of a replaceable battery **22** which provides an output voltage. The battery **22** may be rechargeable. The security device **12.1** also comprises a position determining device **24** which is configured to generate position data and the processor **14** is configured to incorporate the position data into the first message.

The sensor element **13** may comprise any one of: a) an intrusion sensor, such as a passive infrared sensor, a contact sensor for a door or window, photoelectric beam, a panic button or other operable button, a magnetic lock sensor, a strike lock sensor; and b) another sensor, such as a temperature sensor, a smoke sensor, a water or moisture sensor/detector etc. These sensors may be normally open or normally closed sensors.

The first message may be an intrusion message comprising data derived from a signal from an intrusion sensor, alternatively the first message may be a warning message comprising data derived from a signal from at least one of the temperature sensor, smoke sensor, water or moisture sensor etc.

In FIG. 1 is shown the security system **10** which comprises a plurality of security devices **12.1** to **12.n**. The security devices **12.1** to **12.n** are similar to one another. The central backend **20** is in communication with each of the security devices **12.1** to **12.n** individually via the IoT communications path **18** and via the Internet **26**. The IoT communications path **18** comprises at least one IoT high site **28** which is connected to the Internet **26** via an IoT network switch **30**.

The IoT transceiver **16** is preferably a narrow-band type transceiver which transmits the first message via the IoT communications path **18**. The IoT communications path **18** is preferably provided by an IoT network for example one of the networks which are marketed under the trademarks 'SIGFOX'™ or 'LoRaWAN'™ which respectively provides a Low-power Wide-area network (LPWAN). The IoT network switch **30** enables connection to the Internet **26** and a webserver (not shown) performs a signal exchange step between the Internet **26** and the backend **20**.

The security system **10** is associated with a first user station **32.1**, where the security devices **12.1** to **12.n** are deployed in distributed locations at the first user station **32.1**. Further similar security systems may be provided for further user stations **32.2** to **32.m**.

The cloud-based controller **19** is preferably hosted on a server **34** at the remote central backend **20**. The backend **20** may be housed at a single physical site or may be distributed over a plurality of distributed physical sites. The backend **20**

hosts a respective cloud-based controller **19.1** to **19.m** associated with the respective security systems at each of the plurality of distributed user stations **32.1** to **32.m**. The systems associated with the user stations **32.1** to **32.m** are similar and hence the system **10** associated user station **32.1** only will be described in further detail below.

The security system **10** further comprises a terminal which is associated with the first user station **32.1**. The terminal is configurable to be in communication with the cloud-based controller **19.1**. The terminal is user operable to configure settings of the cloud-based controller **19.1** via the Internet **26**. The terminal is preferably in the form of a mobile device **36** of a user **38**, the mobile device **36** comprising a processor, an associated memory arrangement and a display (shown in FIG. 3). The system **10** still further comprises a computer application program (app) which is stored in the mobile device memory arrangement and, in use, is executed by the mobile device's processor to present a user interface (UI) **40** on the display of the terminal, and in response to commands entered via the UI **40** to configure the settings of the cloud-based controller **19.1**. Hence, the user **38** is enabled to control the security devices **12.1** to **12.n** via the IoT communications path **18**.

Any one of the plurality of security devices **12.1** to **12.n** may be configured to communicate via its respective IoT transmitter or transceiver **16** with any other one or more of the other security devices **12.1** to **12.n** at the user station **32.1**.

In FIG. 2 is shown the security device **12.1** in more detail. The position determining device **24** is preferably in the form of a Global Positioning System (GPS) device or a Russian Global Navigation Satellite System (GLONASS) device or any other position determining device for determining the position of the security device **12.1** utilizing satellites **42** (shown in FIG. 1) in known manner. The sensor element **13**, processor **14**, transmitter or transceiver **16**, position determining device **24** and power supply **22** may be housed in a single sensor device housing **43** (shown in FIG. 2). The security device **12.1** is hence self-contained and portable or transportable between different locations at user station **32.1** or between user stations **32.1** to **32.m**.

The processor **14** is configured to incorporate the position data into the first message, in the form of an intrusion message, which is transmitted to the backend **20** via the IoT communications path **18**. The position data may be incorporated into the intrusion message repetitively or only at certain predetermined time intervals. A clock (not shown) may be utilised in determining the time intervals. Alternatively, the position data may be transmitted by the IoT transceiver **16** in a so-called "heartbeat message" for example every 10 minutes or at any other suitable time. The "heartbeat message" may in addition comprise data relating to a current output voltage of the battery **22** and data relating to a unique identification (ID) number **44** of the security device **12.1** which uniquely identifies the device on the IoT network **18**. The data relating to the unique ID of the security device **12.1** is preferably pre-stored into a memory arrangement **46** associated with the security device **12.1**. The user **38** may for example be alerted of a low battery voltage of one of the security devices **12** via the app on the mobile device **36**.

In FIG. 3 is shown the mobile device **36** of the user **38** in more detail. The user **38** is prompted by the app to register at the backend **20**, each of the security devices **12.1** to **12.n** which are associated with the user station **32.1** of the user **38**.

Referring again to FIG. 1, the backend **20** comprises the cloud-based controller **19.1** which is hosted by the server **34**. The backend **20** also comprises a database **48**. The above registration by the user via the UI **40** on mobile device **36** involves entering into and storing of data relating to the user **38**, data relating to the user station **32.1** and data relating to the unique ID's of the first user station's associated security devices **12.1** to **12.n** on database **48** at backend **20**. The registration process may comprise scanning a code associated with each respective security device **12.1** to **12.n** such as a bar code or a "QR code"™ when the user **38** is logged into a user account via the app. It will be appreciated that a plurality of users **38.1** to **38.m** may be registered in this manner at the backend **20**, for respective user stations **32.1** to **32.m**.

In use, the user **38** purchases one or more of the security devices **12.1** to **12.n**, installs them at the user station **32.1** and registers them at the backend **20**. Referring to FIG. 3, the user is enabled to configure settings of the cloud-based controller **19.1** via the UI **40**. Hence, the user **38** is enabled to control the security devices **12.1** to **12.n** via the app and via the IoT communications path **18**. The user **38** may utilise the UI **40** to assign each of the security devices **12** an appropriate name, for example kitchen **50**, bedroom **52**, garage **54**, driveway **56**, etc. The user **38** is also enabled to activate or deactivate (by-pass) each of the security devices **12.1** to **12.n** individually via the UI **40**. Some of the security devices **12.1** to **12.n** may also be grouped together, for example security devices **12.1** to **12.3** may be assigned to the kitchen **50**, security devices **12.4** to **12.6** to the bedroom **52** etc. A user may use buttons **58**, **60** and **62** on the UI **40** to cause the cloud-based controller **19.1** to operate in a user selectable ARM mode, a DISARM mode and a STAY mode, subject to the user bypass selections referred to above.

The backend **20** may also be configured to alert the user **38** via the app when one of the security devices **12.1** to **12.n** is activated or when the GPS device **24** provides position data indicating that the security device **12.1** is located outside a geographical coverage area of an operator of the backend **20**.

It will be appreciated that there are many variations in detail on the security device, security system, and associated method without departing from the scope and spirit of this disclosure.

The system **10** provides multiple redundancies and robustness in that when one of the security devices such as security device **12.1** has limited or no connectivity to the IoT network **18**, the respective device is enabled to transmit the intrusion message to one of the other security devices **12.2** to **12.n** which device then forwards the intrusion message to the backend **20** via the IoT communications path **18**.

Apart from a first message in the form of an intrusion message, the above other sensors may output signals which may be used to generate a first signal in the form of a warning message. For example, leaks of a water heater (also referred to as a "geyser") may be monitored utilising a water detector. In this case, the processor **14** may be configured, in the event of a water leak, to transmit to the backend **20** via the IoT communications path **18**, a warning or other message comprising data relating to a detected leak and position data. It will be appreciated that these other messages may comprise data associated with any of the other sensors, with or without position data.

A further position determining device (not shown) may be associated with the mobile device **36** and the system **10** may be configured such that when the mobile device **36** is located in, alternatively, outside a predefined geographical region

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(sometimes referred to as “geo-fencing”), one or more of the security devices **12.1** to **12.n** and/or modes may automatically be enabled or disabled by the cloud-based controller **19.1**.

The invention claimed is:

1. A security system comprising:

at least one security device at a first user station, the at least one security device comprising:

a portable housing;

a local power supply for providing an output voltage; a sensor element;

a processor connected to the sensor element for generating a first message in response to a signal from the sensor element; and

an Internet of things transmitter connected to the processor;

a cloud-based controller for the security system, the cloud-based controller being hosted on a server at a central backend that is positioned at a location remote from the user station, the cloud-based controller being in data communication with the at least one security device via an Internet of things communications path, the cloud-based controller having an arm mode of operation and a disarm mode of operation;

a terminal associated with the first user station, the terminal being configured to be in communication with the cloud-based controller to cause the cloud-based controller to operate in a user selectable one of the arm mode of operation or the disarm mode of operation, and the terminal being operable to configure settings of the cloud-based controller to cause the cloud-based controller to act on the first message in accordance with one or more settings pre-stored on the cloud-based controller,

wherein:

the Internet of things transmitter is a narrow-band transmitter configured to transmit the first message directly to a high site, external of the user station, of a Low-power Wide-area network providing the Internet of things communications path;

the local power supply comprises at least one replaceable battery;

the sensor element, the processor, the narrow-band transmitter, and the at least one replaceable battery are housed in the portable housing of the at least one security device; and

the at least one security device is wholly self-powered and self-contained, as well as portable and transportable between different locations at the user station and between the user station and another user station.

2. The security system as claimed in claim **1**, further comprising a memory arrangement connected to the processor and for storing a unique identification number (ID) associated with the device and for use in communications via the network.

3. The security system as claimed in claim **1**, wherein the sensor element comprises any one of: a passive infrared sensor, a contact sensor for a door or window, a photoelectric beam, a panic button or other user operable button, a magnetic lock sensor, a strike lock sensor, a temperature sensor, a smoke sensor and a water or moisture sensor.

4. The security system as claimed in claim **1**, wherein the processor is configured to incorporate, into the first message, data relating to a current output voltage of the power supply.

5. The security system as claimed in claim **1**, further comprising a position determining device which is connected to the processor.

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6. The security system as claimed in claim **5**, wherein the processor is configured to incorporate, into the first message, position data as determined by the position determining means.

7. The security system as claimed in claim **1**, further comprising an Internet of things receiver connected to the processor and for receiving a second message from the cloud-based controller via the Internet of things network.

8. The security system as claimed in claim **1**, wherein the terminal is a mobile device comprising a processor, an associated memory arrangement, and a display.

9. The security system as claimed in claim **8**, further comprising a computer application stored in the memory arrangement, the computer application being, in use, executed by the processor to present a user interface on the display of the terminal and, in response to commands entered via the user interface, to configure the settings of the cloud-based controller.

10. The system as claimed in claim **1**, further comprising a plurality of the security devices which are deployable at distributed locations at the first user station.

11. A method of protecting a user station, the method comprising:

at a cloud-based controller for a security system associated with the user station, receiving from a user terminal associated with the user station a command to cause the cloud-based controller to operate in a user selectable arm mode of operation or a user selectable disarm mode of operation and receiving from a security device located at the user station a first message, the first message being transmitted by a narrow-band Internet of things transmitter directly to a high site, external of the user station, of a Low-power Wide-area network and via the network to the cloud-based controller; and causing the cloud-based controller to act on the first message in accordance with settings which are pre-stored on the cloud-based controller,

wherein:

the cloud-based controller is hosted on a server at a central backend that is positioned at a location remote from the user station;

the security device comprises a local power supply comprising at least one replaceable battery; and

the narrow-band Internet of things transmitter is housed with a sensor element, a processor, and the at least one replaceable battery in a portable housing, so that the security device is wholly self-powered and self-contained, as well as portable and transportable between different locations at the user station and between the user station and another user station.

12. A security device comprising:

a portable housing;

a local power supply for providing an output voltage;

a sensor element;

a processor connected to the sensor element for generating a first message in response to a signal from the sensor element; and

an Internet of things transmitter connected to the processor,

wherein:

the Internet of things transmitter is a narrow-band transmitter configured to transmit the first message directly to a high site, external of the user station, of a Low-power Wide-area network and via the network to a cloud-based controller of a security system of which the security device forms part;

the local power supply comprises at least one replaceable battery; and

the sensor element, the processor, the narrow-band transmitter, and the at least one replaceable battery are housed in the portable housing, so that the at least one security device is wholly self-powered and self-contained, as well as portable and transportable between different locations at the user station and between the user station and another user station.

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